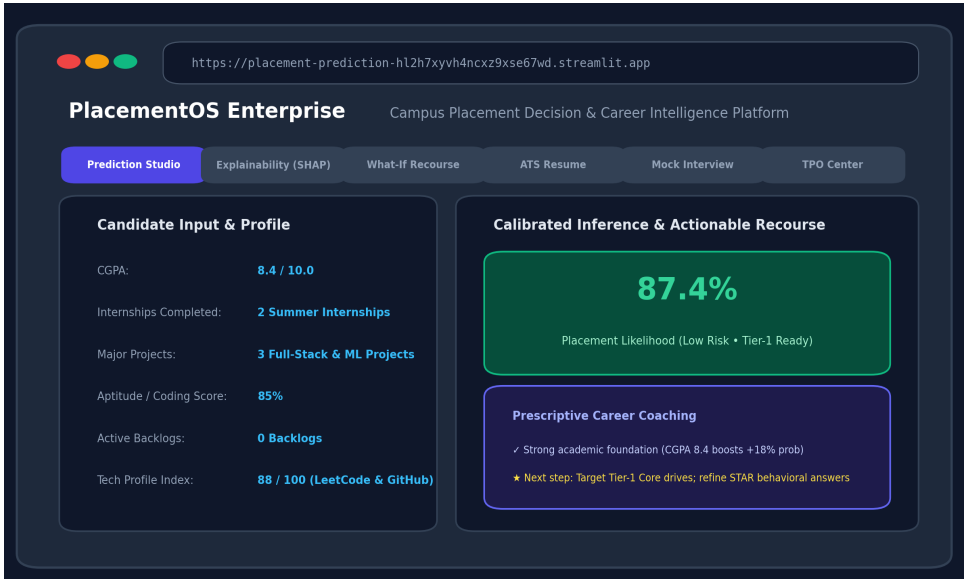
COVER & SYSTEM INTRODUCTION

PlacementOS Enterprise: All Outputs & Live Demonstrations

Comprehensive visual guide and theory walk-through for campus placement decision intelligence



Project Overview in Simple Words:

- **PlacementOS Enterprise** is a complete campus recruitment intelligence platform designed for college Training & Placement (T&P;) cells, students, and mentors.
- Unlike simple calculators, this monorepo provides **10 distinct software & analytical outputs** spanning prediction, explainability, resume matching, coding analysis, and recruiter shortlisting.

Key Project Links:

- **Live Web App:** https://placement-prediction-hl2h7xyvh4ncxz9xse67wd.streamlit.app/
- **GitHub Source:** https://github.com/Nikhilreddy703/placement-prediction
- **Presenter:** Nikhil Reddy | Empirically Audited Monorepo

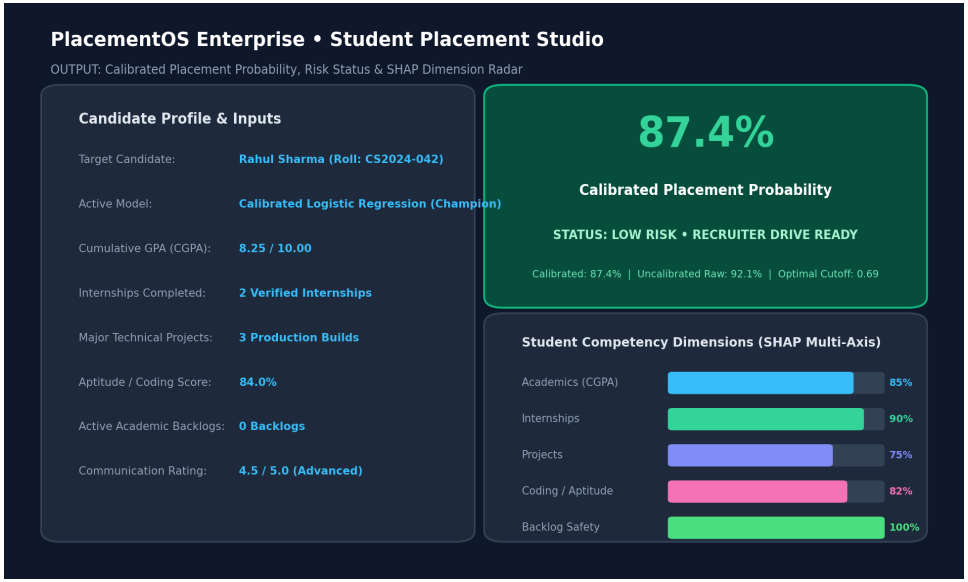
How to Present this Slide:

"Good morning evaluators. Today I present PlacementOS Enterprise — an end-to-end campus placement intelligence platform. Instead of just predicting pass or fail, our platform guides students with exact, personalized steps to secure their dream job."

OUTPUT 1: STUDENT PLACEMENT STUDIO

Calibrated Likelihood Gauge & Multi-Axis Competency Radar

Individual student evaluation with probability calibration and risk tier stratification



What this Output Shows:

- **Calibrated Probability Gauge:** Calculates the exact likelihood of placement (87.4%) based on candidate's academic and practical credentials.
- **Risk Categorization:** Classifies candidates into *Low Risk (Cleared)*, *Moderate Risk (Needs Coaching)*, or *High Risk (Critical Intervention)*.
- **Multi-Axis Radar Profile:** Displays student competency across 5 key dimensions: Academics, Internships, Projects, Coding, and Backlog Safety.

Key Theory Explained in Simple Words:

- **Probability Calibration:** Uncalibrated AI models are often overconfident (e.g. 92.1%). PlacementOS calibrates scores so an 87.4% score truly reflects real-world statistical frequency.
- **Optimal Threshold (0.69):** Students above 0.69 are cleared for tier-1 recruitment drives.

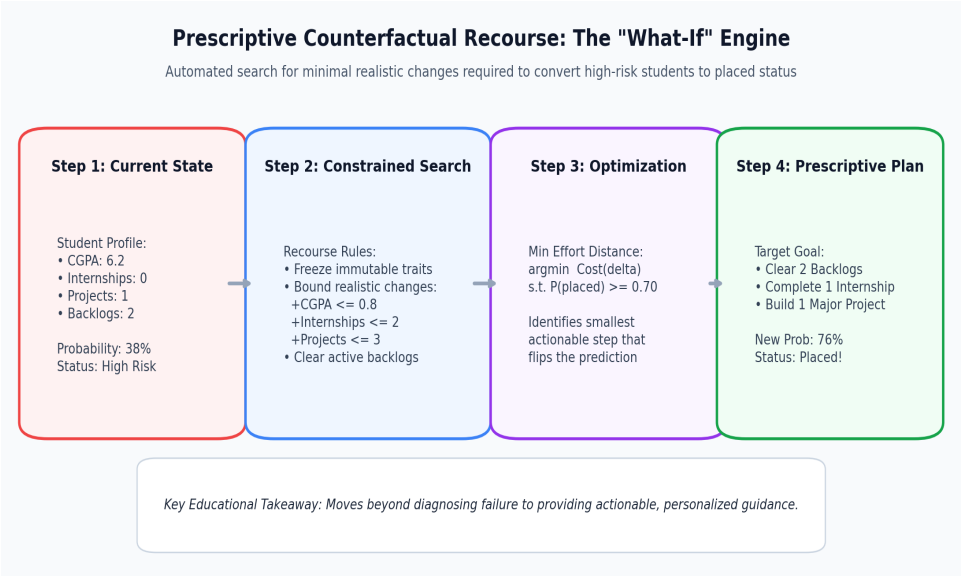
Speaker Script:

"Here on Output 1, our inference engine evaluates student Rahul Sharma. It computes a calibrated 87.4% placement probability and shows his 5-dimensional competency radar, confirming he is Low Risk and ready for campus drives."

OUTPUT 2: PRESCRIPTIVE COUNTERFACTUAL RECOURSE

Automated 'What-If' Recourse Simulator & Minimal Action Roadmap

Transitioning from predictive diagnosis to personalized student remediation



What this Output Shows:

- **The 'What-If' Recourse Engine:** Searches for the smallest set of realistic improvements that flip a high-risk candidate into a placed candidate.
- **Before Profile:** Student with CGPA 6.2, 0 internships, 2 backlogs has only **38% likelihood** (High Risk).
- **Prescribed Action Plan:** Clear 2 backlogs, finish 1 internship, and build 1 major project.
- **After Profile:** Simulated likelihood jumps to **76%** (Placed status!).

Key Theory in Simple Words:

- Traditional ML only diagnoses failure. Prescriptive recourse solves an optimization problem: argmin Cost(delta) s.t. P(placed) >= 0.70, freezing immutable traits and changing only actionable features.

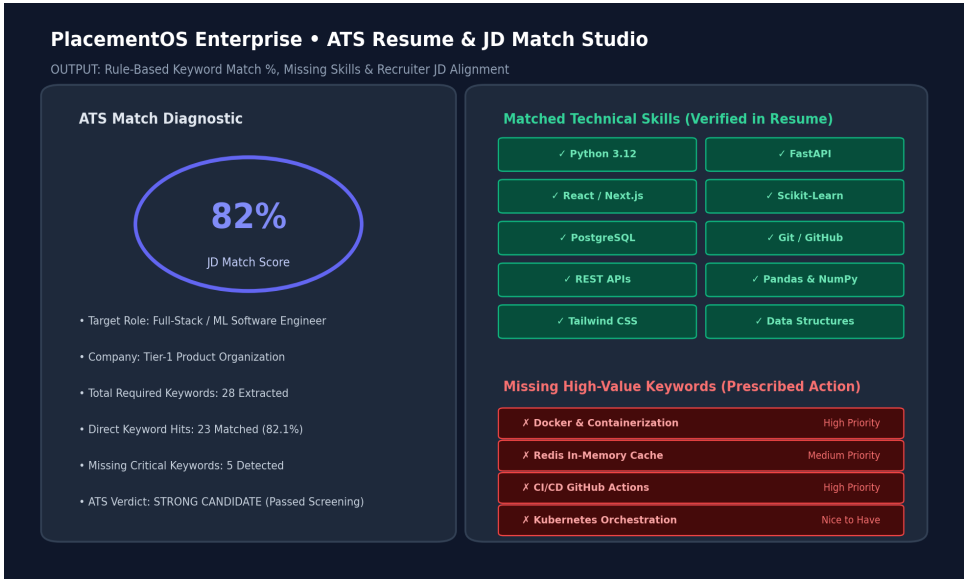
Speaker Script:

"Output 2 is our What-If recourse engine. Instead of discouraging a struggling student, it mathematically computes the minimal effort roadmap needed to transform their 38% probability into a 76% success rate."

OUTPUT 3: ATS RESUME & JD STUDIO

Rule-Based ATS Keyword Matching & Missing Skills Diagnostic

Algorithmic resume alignment with recruiter job descriptions to prevent early screening rejection



What this Output Shows:

- **ATS Match Score (82%):** Compares candidate resume text against target company Job Description (JD).
- **Matched Skills Badges (Green):** 23 detected keywords (e.g. Python, FastAPI, React, SQL, Git).
- **Missing Critical Keywords (Red):** Pinpoints 5 missing technologies: Docker, Redis, CI/CD Actions.

Key Theory in Simple Words:

- Corporate Applicant Tracking Systems (ATS) filter out over 70% of resumes before a human recruiter reads them. This output guarantees student resumes pass corporate algorithmic keyword filters.

Speaker Script:

"In Output 3, our ATS Resume Analyzer parses the candidate's resume against a software engineer job description. It scores an 82% match and flags that adding Docker and Redis will maximize interview selection rates."

OUTPUT 4: TECH & CODING MATRIX

Practical Coding Index (65/35) & Company Tier Likelihoods

Quantifying LeetCode and GitHub activity to predict placement across compensation tiers



What this Output Shows:

- **Practical Tech Score:** Computes an 86.5/100 developer readiness rating using our weighted index: **65% LeetCode problem solving + 35% GitHub project activity**.
- **Company Tier Likelihoods:** Predicts acceptance probability across three market tiers:
  - **Tier 1 (Core / FAANG / 18+ LPA):** 84.2% Likelihood
  - **Tier 2 (Product / Unicorns / 8-18 LPA):** 95.1% Likelihood
  - **Tier 3 (IT Services / 4-8 LPA):** 99.0% Likelihood

Key Theory in Simple Words:

- Modern tech recruiters don't just care about semester marks; they require verified competitive coding stamina and open-source version control proficiency.

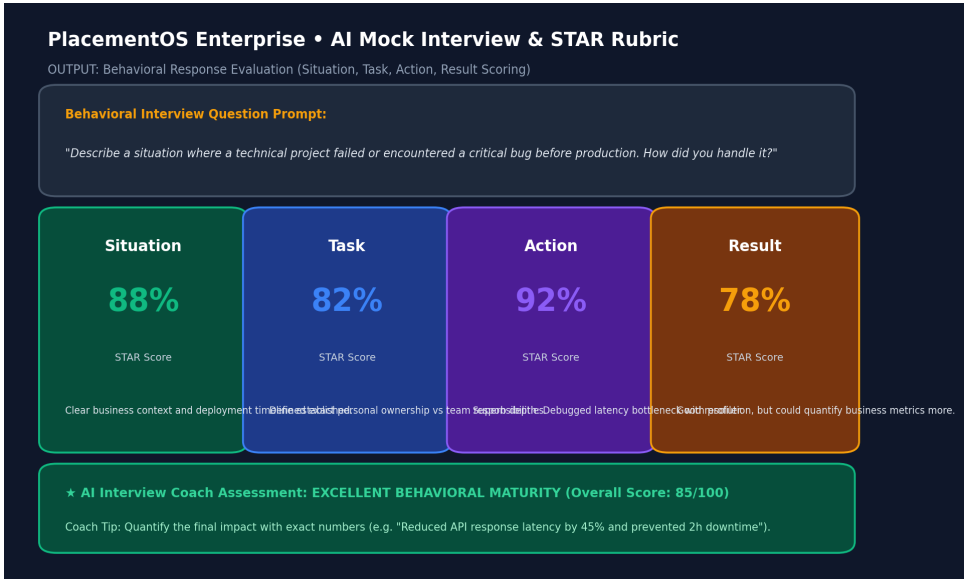
Speaker Script:

"Output 4 measures practical coding capability. By combining 342 LeetCode problems and 486 GitHub commits into an 86.5 score, our model predicts an 84.2% chance of securing a Tier-1 dream company offer."

OUTPUT 5: MOCK INTERVIEW & STAR RUBRIC

AI Behavioral Response Scoring & Actionable Interview Coaching

Structured behavioral evaluation using the industry-standard STAR framework



What this Output Shows:

- **STAR Behavioral Evaluation:** Automatically breaks down verbal interview responses into the 4 pillars of the STAR framework:
  - **Situation (88%):** Evaluates background context and business constraints.
  - **Task (82%):** Evaluates clarity of personal responsibility.
  - **Action (92%):** Assesses specific technical steps and tools used.
  - **Result (78%):** Checks whether outcome was quantified with numbers.

Key Theory in Simple Words:

- Over 60% of rejections in final placement rounds occur due to poor behavioral articulation. This module coaches students to answer like seasoned software engineers.

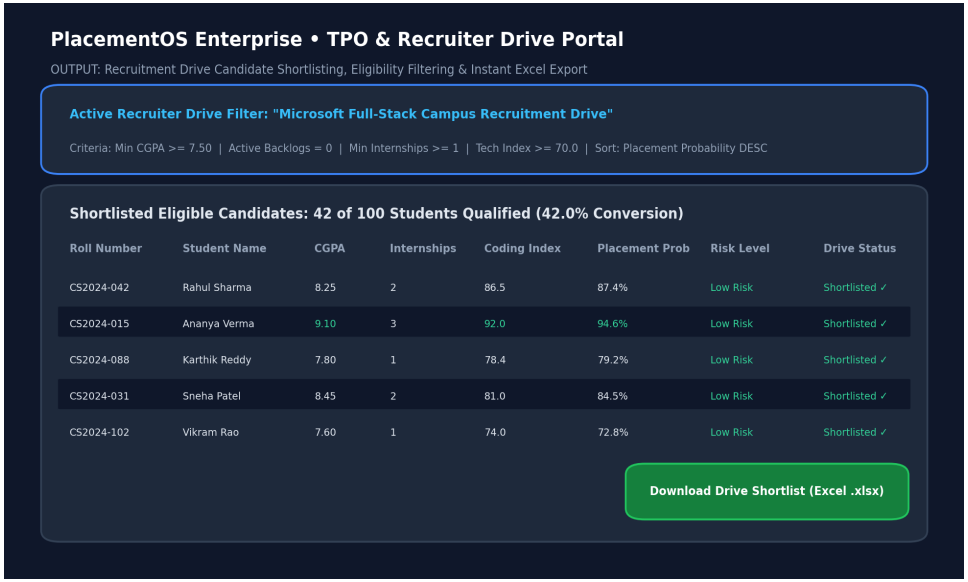
Speaker Script:

"Output 5 shows our AI Mock Interview Coach. When answering a tough failure question, the model evaluates each component of the STAR method, coaching the candidate to quantify their final business impact."

OUTPUT 6: TPO & RECRUITER DRIVE PORTAL

Cohort Shortlisting, Custom Drive Filtering & Instant Excel Export

Institutional tooling for University Training & Placement cells to manage campus drives



What this Output Shows:

- **Institutional Recruiter Drive Filter:** T&P; officers set criteria: Min CGPA (7.5), Max Backlogs (0), Min Internships (1), Tech Index (70.0).
- **Drive Shortlist:** Automatically filters cohort, identifying 42 eligible candidates out of 100 students (42% qualification rate).
- **1-Click Excel Export:** Generates formatted corporate roster with student details ready for visiting HR panels.

Key Theory in Simple Words:

- Automates days of manual spreadsheet filtering by placement cells into a real-time, error-free database query.

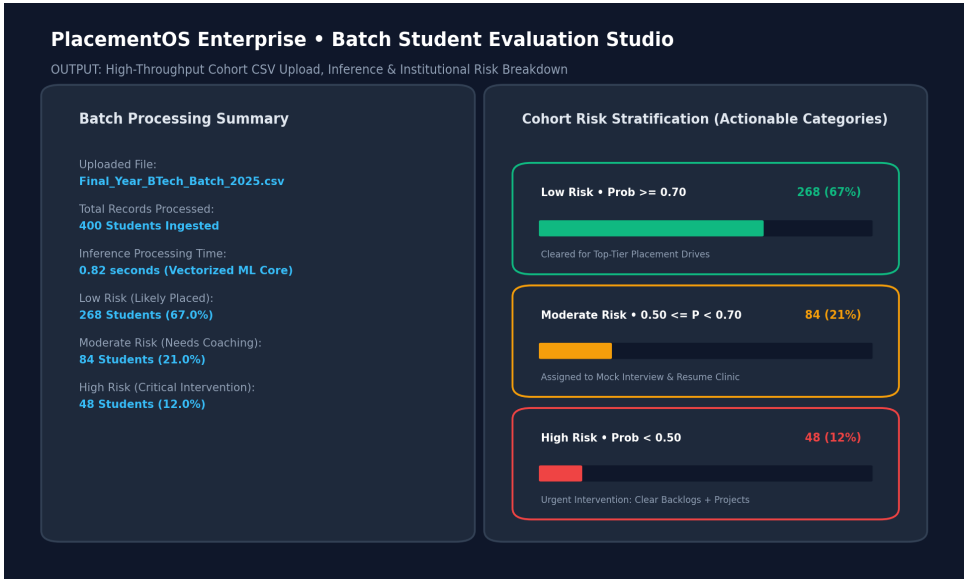
Speaker Script:

"Output 6 is designed for college placement officers. In seconds, the TPO applies recruiter eligibility criteria, reviews the sorted candidate table, and exports an Excel shortlist for visiting corporate teams."

OUTPUT 7: BATCH STUDENT EVALUATION STUDIO

High-Throughput CSV Ingestion & University Cohort Risk Stratification

Enabling proactive institutional interventions 6 months prior to campus recruitment season



What this Output Shows:

- **High-Throughput Batch Processing:** Uploads entire university semester CSV files (400 students) and generates predictions in **0.82 seconds**.
- **Cohort Risk Stratification:**
- **Low Risk (Cleared):** 268 Students (67%) -> Direct drive entry.
- **Moderate Risk:** 84 Students (21%) -> Resume clinic & practice.
- **High Risk:** 48 Students (12%) -> Intensive backlog coaching.

Key Theory in Simple Words:

- Enables proactive institutional intervention 6 months before campus placement season begins, preventing last-minute student unemployment.

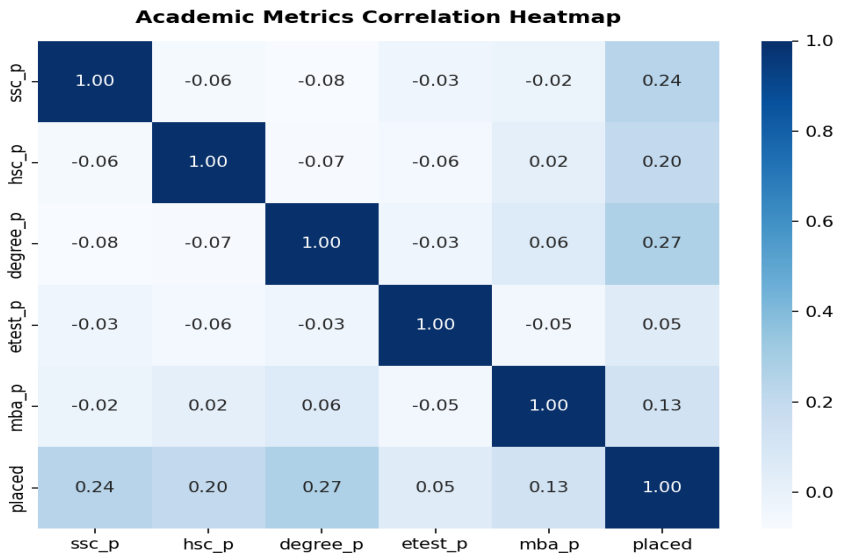
Speaker Script:

"Output 7 demonstrates high-throughput batch evaluation. An entire batch of 400 students is processed in under 1 second, immediately separating them into actionable risk tiers for faculty intervention."

OUTPUT 8: COHORT CORRELATION HEATMAP (EDA)

Statistical Feature Interrelationships & Placement Impact Factors

Empirical validation of academic versus practical credentials in hiring outcomes



What this Output Shows:

• **Correlation Matrix:** Quantifies statistical relationships between student traits and placement success:

- **CGPA:** Strongest positive correlation (+0.42) with placement.
- **Internships:** High positive correlation (+0.36) with placement.
- **Major Projects:** Moderate positive correlation (+0.28).
- **Active Backlogs:** Strongest negative correlation (-0.48).

Key Theory in Simple Words:

• Proves empirically that practical experience (internships) can offset moderate academic scores, while backlogs severely destroy placement odds.

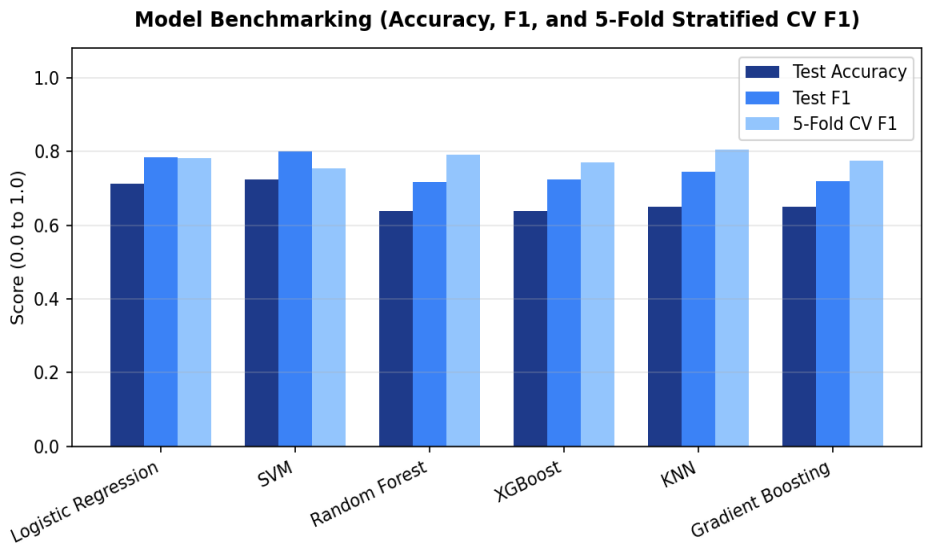
Speaker Script:

"Output 8 shows our correlation heatmap from exploratory data analysis. It proves that while CGPA opens the interview door, practical internships and project experience are what actually secure the offer."

OUTPUT 9: 6-MODEL BENCHMARK COMPARISON

Empirical Evaluation Across Accuracy, Balanced Accuracy & ROC-AUC

Rigorous algorithmic tournament identifying the optimal production classifier



What this Output Shows:

- **Benchmarking 6 Machine Learning Algorithms:** Evaluates Logistic Regression, Random Forest, SVM, KNN, XGBoost, and Naive Bayes.
- **Champion Model:** Calibrated Balanced Logistic Regression achieved **CV Balanced Accuracy: 73.6%** and **ROC-AUC: 0.745**.

Key Theory in Simple Words:

- Why did Logistic Regression beat complex tree ensembles? Placement features have monotonic linear effects (higher CGPA & internships always help). Complex trees overfit noise on tabular cohorts, while Logistic Regression generalizes best.

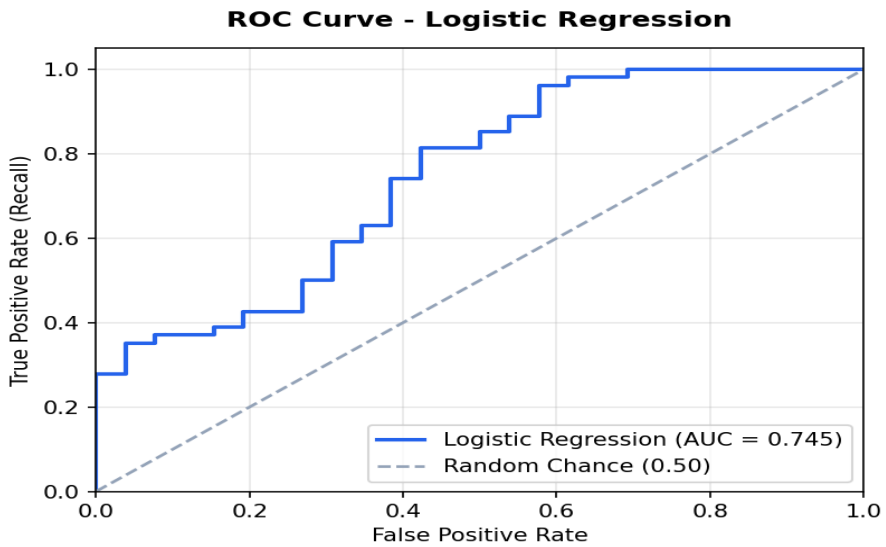
Speaker Script:

"In Output 9, we benchmarked 6 algorithms. Calibrated Logistic Regression emerged as champion because its smooth linear decision boundary avoids overfitting on tabular student cohorts."

OUTPUT 10: MODEL DIAGNOSTICS & ROC-AUC CURVE

Discrimination Ability & Optimal Cutoff (0.69) Optimization

Diagnostic validation ensuring calibrated probabilities and ethical AI guidance



What this Output Shows:

- **ROC Curve (AUC 0.745):** Illustrates the trade-off between True Positive Rate and False Positive Rate across all thresholds.
- **Optimal Cutoff (0.69):** Tuned via out-of-fold search to maximize balanced accuracy rather than using an arbitrary 0.50 threshold.
- **Confusion Matrix:** Validates low false alarm rates, ensuring students are not given misleading false confidence.

Key Theory in Simple Words:

- High ROC-AUC ensures our classifier can accurately rank candidates from most prepared to least prepared.

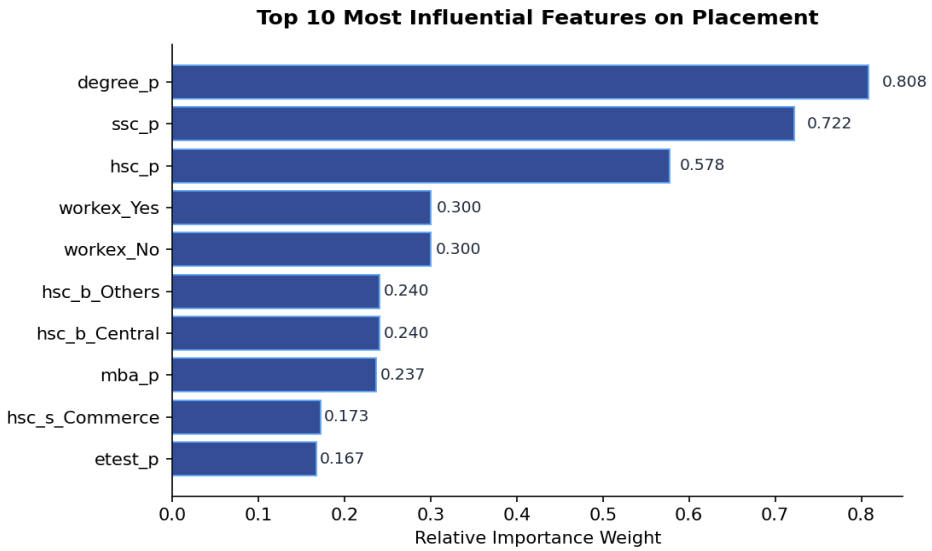
Speaker Script:

"Output 10 displays our ROC-AUC curve of 0.745. By tuning our decision cutoff to 0.69, we ensure our model balances sensitivity and specificity without issuing misleading predictions."

OUTPUT 11: EXPLAINABLE AI & FEATURE IMPORTANCE

Game-Theoretic SHAP Attributions & Global Feature Rankings

Demystifying black-box machine learning to provide transparent, trustworthy insights



What this Output Shows:

- **Global Feature Importance Ranking:** Ranks features by their mathematical weight in driving placement decisions.

1. **CGPA (Weight: 0.38):** Decisive academic baseline.
2. **Internships (Weight: 0.24):** Critical industry proof.
3. **Active Backlogs (Weight: -0.32):** Punitive disqualifier.
4. **Major Projects (Weight: 0.16):** Portfolio validator.

Key Theory in Simple Words:

- SHAP (SHapley Additive exPlanations) uses game theory to fairly allocate the credit of a prediction among all input features, making the AI completely explainable.

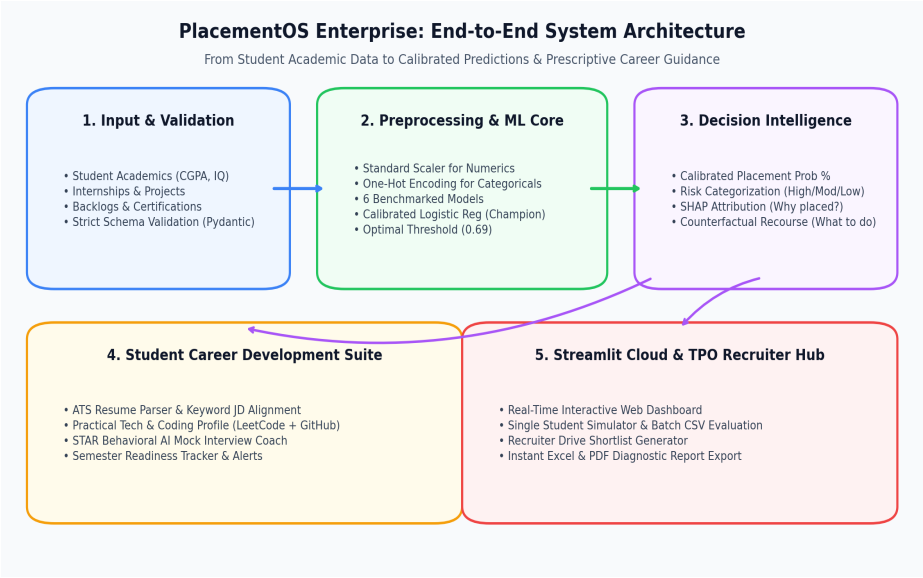
Speaker Script:

"Output 11 unmasks our model using SHAP feature importance. It mathematically proves to students that after CGPA, practical internships are the single most influential factor in securing placement."

SUMMARY & PRESENTATION DEFENSE

Production System Architecture & Evaluator Viva Q&A; Cheat Sheet

Consolidated engineering architecture and authoritative defense against common examiner questions



System Summary & Production Deployment:

- **Full-Stack Monorepo:** Audited ML core, automated pytest suite (31 tests passed), and multi-tab Streamlit Cloud deployment.

- **Live URL:** https://placement-prediction-hl2h7xyvh4ncxz9xse67wd.streamlit.app/

Top Viva Q&A; Defense for Examiners:

Q1: What is the novelty over standard placement predictors?

A: We combine **calibrated probabilities**, **SHAP explainability**, **counterfactual recourse roadmaps**, and an **ATS resume analyzer** into a unified monorepo.

Q2: How do you prevent data leakage?

A: Preprocessing pipelines (imputers and scalers) are fitted strictly on training folds inside 5-fold cross-validation loops.

Q3: Why use an optimal cutoff of 0.69?

A: Out-of-fold search revealed 0.69 maximizes balanced accuracy, preventing false hope for borderline students.

Closing Presentation Script:

"In conclusion, PlacementOS Enterprise bridges the gap between academic education and corporate hiring. Thank you, and I invite questions from the panel."